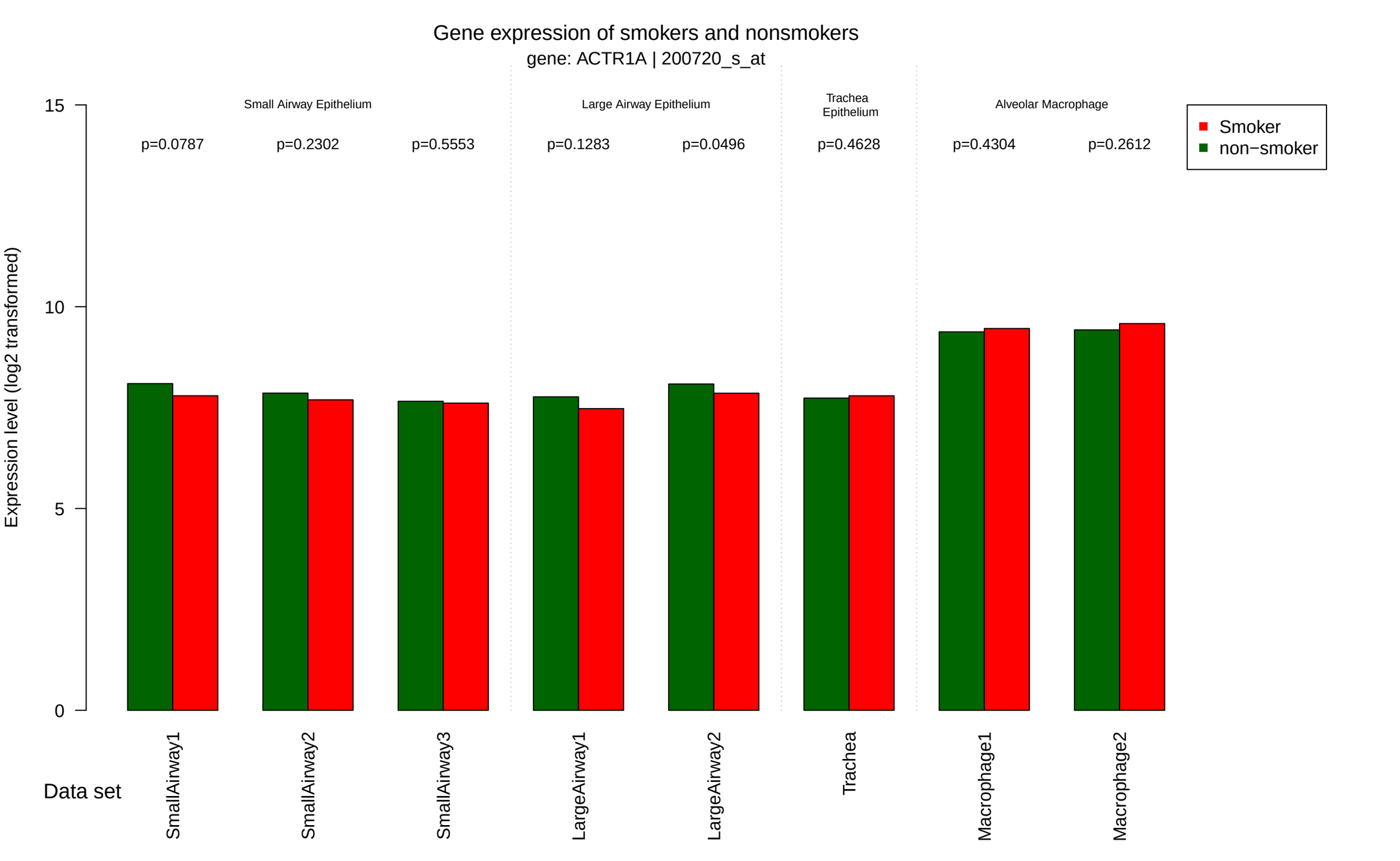
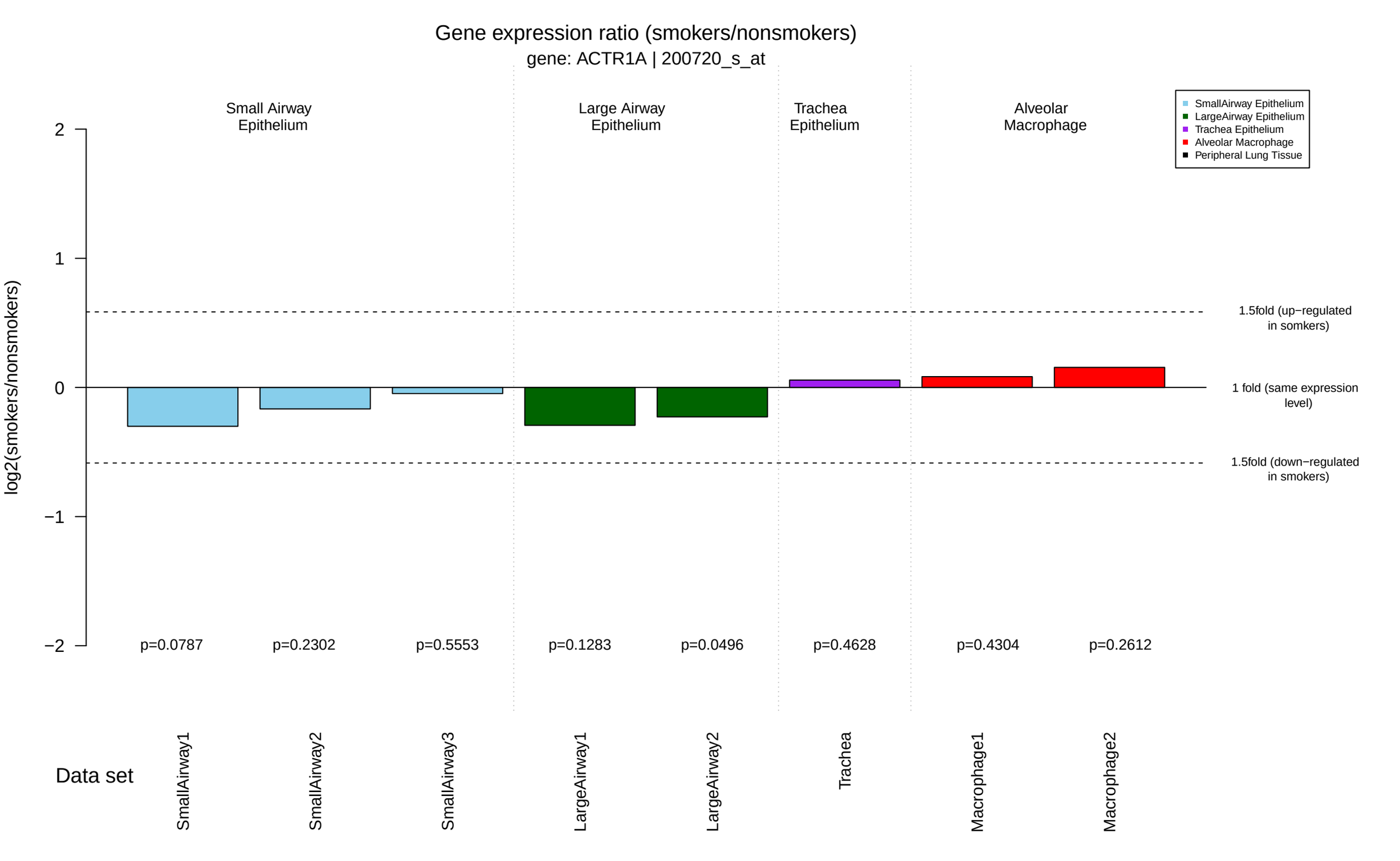
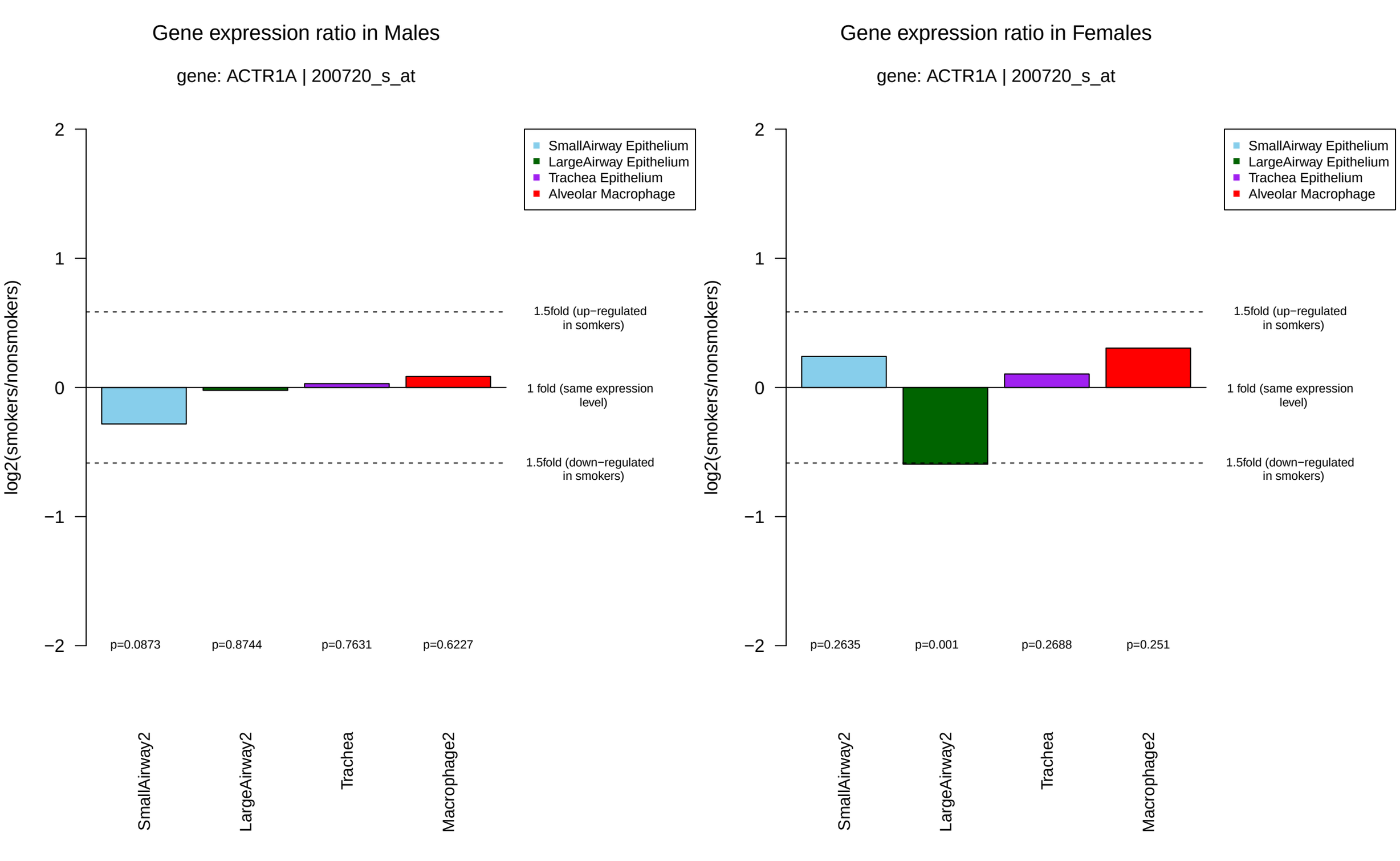
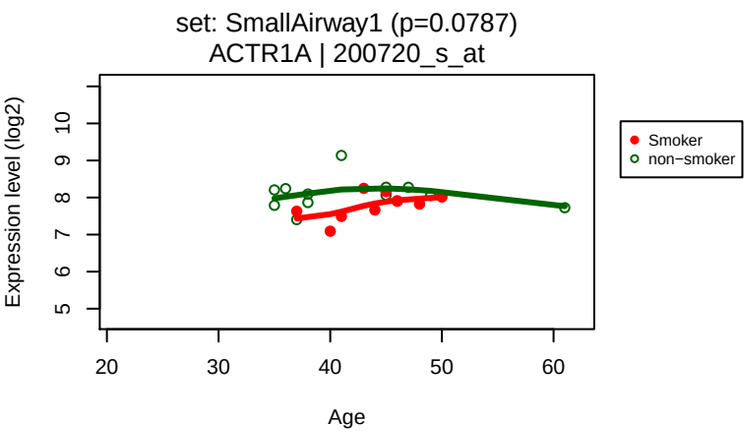
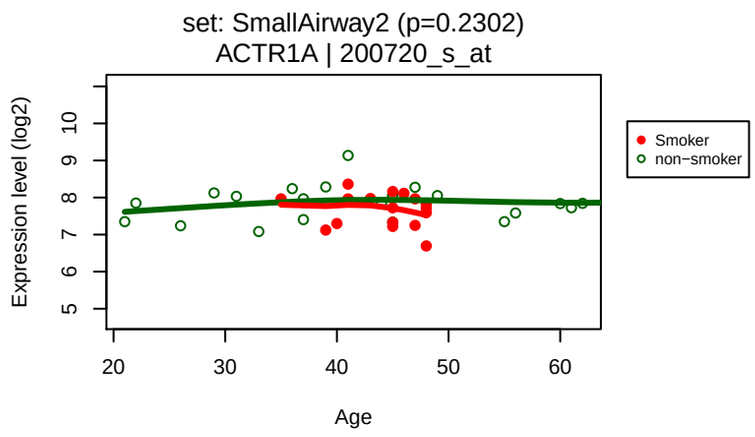
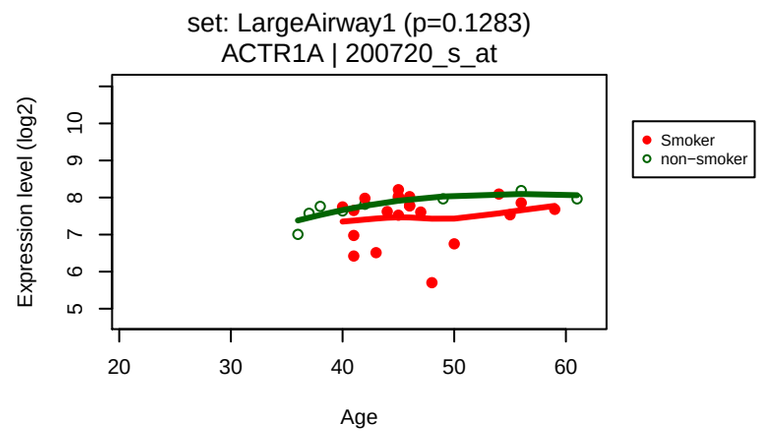

Gene expression of smokers and nonsmokers
gene: ACTR1A | 200720_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0787
p=0.2302
p=0.5553
p=0.1283
p=0.0496
p=0.4628
p=0.4304
p=0.2612
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

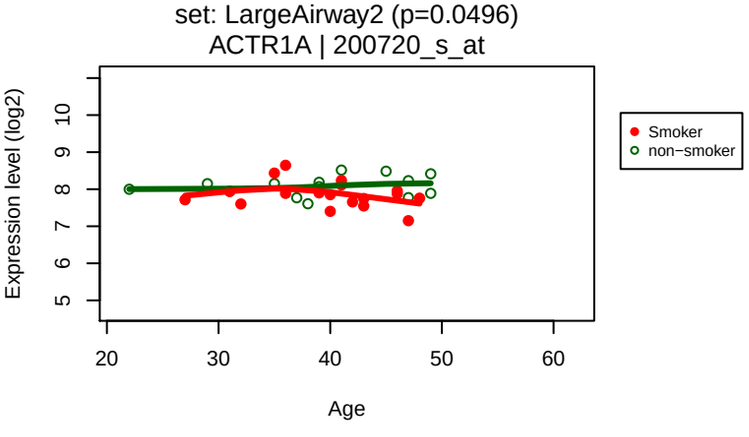

Gene expression ratio (smokers/nonsmokers)
gene: ACTR1A | 200720_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0787
p=0.2302
p=0.5553
p=0.1283
p=0.0496
p=0.4628
p=0.4304
p=0.2612
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

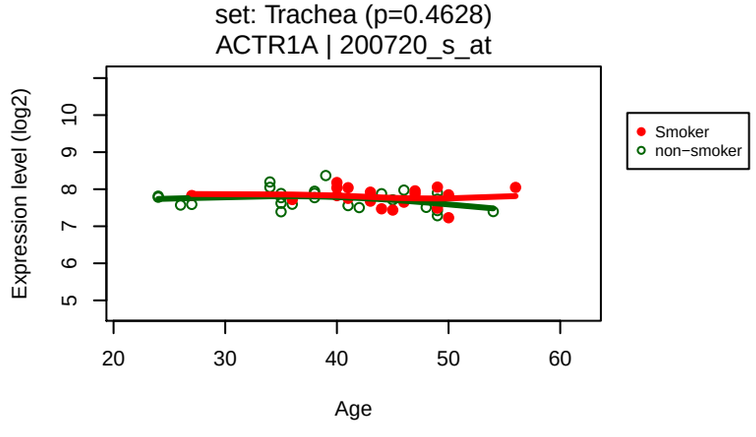

Gene expression ratio in Males
gene: ACTR1A | 200720_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0873
p=0.8744
p=0.7631
p=0.6227
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR1A | 200720_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2635
p=0.001
p=0.2688
p=0.251
SmallAirway2
LargeAirway2
Trachea
Macrophage2

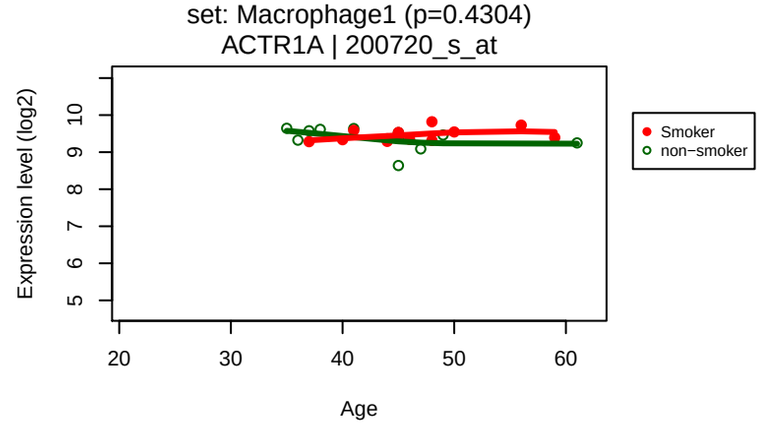

set: SmallAirway1 (p=0.0787)
ACTR1A | 200720_s_at
Expression level (log2)
Age
Smoker
non−smoker

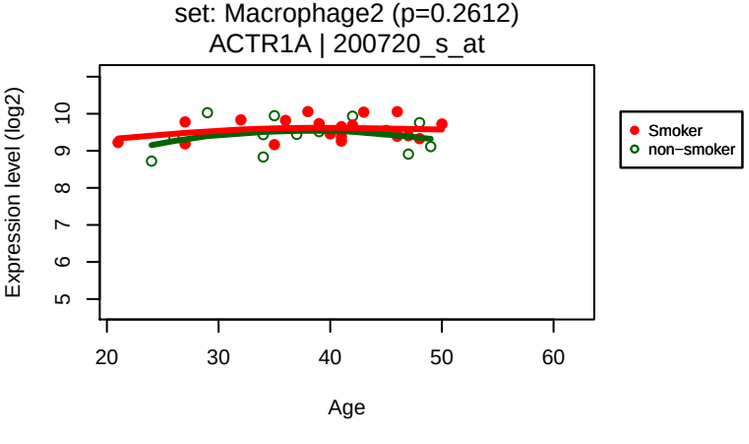

set: SmallAirway2 (p=0.2302)
ACTR1A | 200720_s_at
Expression level (log2)
Age
Smoker
non−smoker

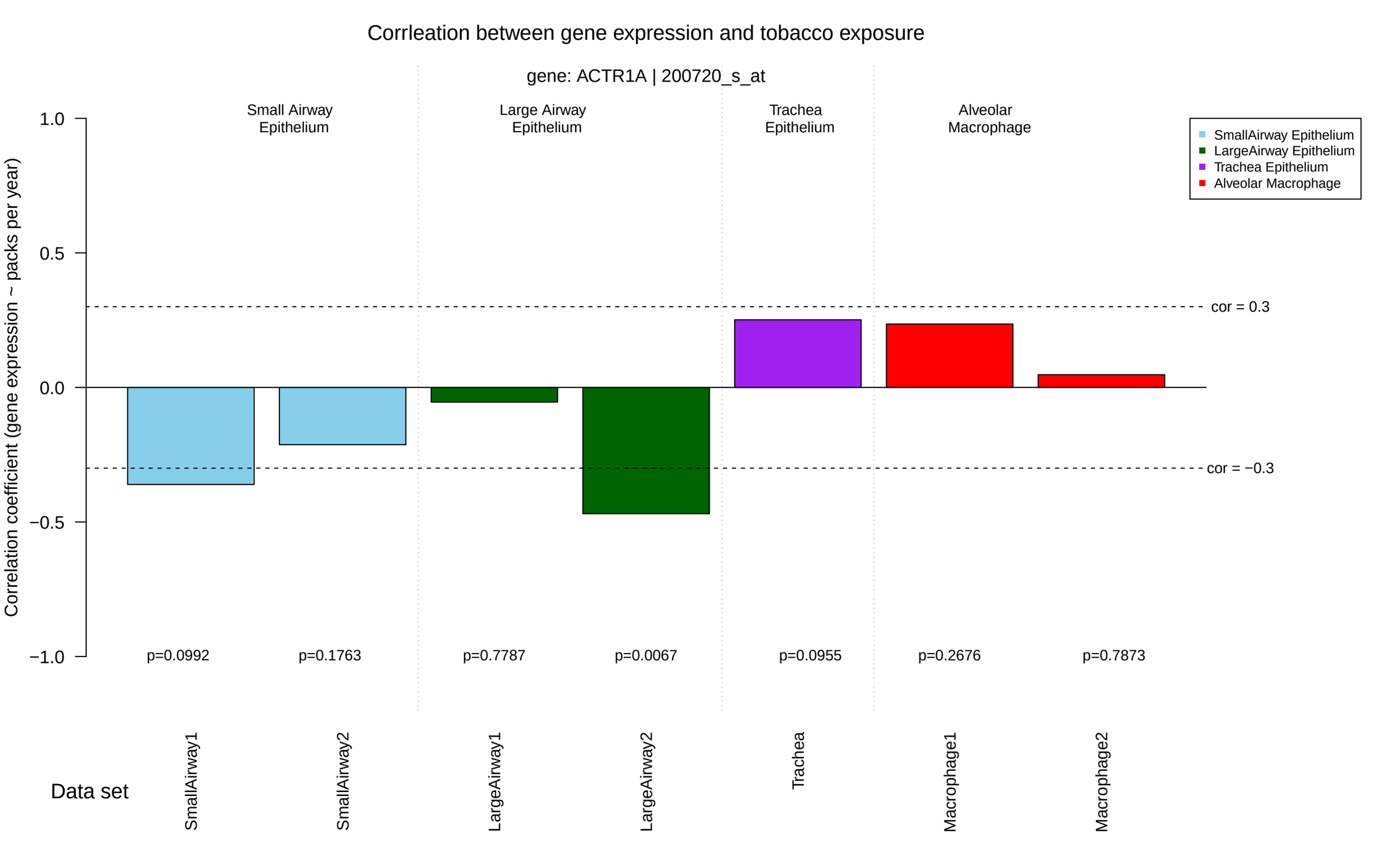

set: LargeAirway1 (p=0.1283)
ACTR1A | 200720_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0496)
ACTR1A | 200720_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4628)
ACTR1A | 200720_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4304)
ACTR1A | 200720_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2612)
ACTR1A | 200720_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR1A | 200720_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0992
p=0.1763
p=0.7787
p=0.0067
p=0.0955
p=0.2676
p=0.7873
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2